

ADVAITH MALKA

(571) -529-2877 • Aldie, VA • advaithm06@gmail.com • Current Top Secret Clearance
linkedin.com/in/advaithmalka • github.com/advaithmalka • advaithmalka.com

EDUCATION

Virginia Tech – Blacksburg, VA

May 2027

Bachelor of Science – Computer Science

GPA: 3.96/4.0 (Dean's List)

Core Courses: Data Structures & Algorithms, Computer Systems, Cloud Software Development, Introduction to AI

EXPERIENCE

RTX, Cybersecurity and AI Engineering Intern

May 2025 – August 2026 | Cedar Rapids, IA

- Built **Neo4j** knowledge graph and **AI Agent** pipeline to generate RMFs, reducing compliance writing time by **50%**.
- Developed **AI agent** with **RAG** for **context-aware** internal-library code generation, speeding development by **70%**.
- Built **LLM-guided fuzzer** that compared **GRM/DUT** outputs and found **20%** more unique crashes than **AFL++**.
- Containerized **3** RAG services with Docker Compose, cutting setup **~80%** and saving **~2** engineer-hours/deployment.

ROLE Lab, Undergraduate Research Assistant | [GitHub](#)

August 2025 – May 2026 | Blacksburg, VA

- Fine-tuned an LLM to emit **uncertainty tokens**, triggering **RAG** only below thresholds and cutting retrieval **~30%**.
- Engineered a **RAG evaluation** pipeline using **vLLM** and **ElasticSearch** to benchmark LLMs on **30K+** queries.
- Outperformed AdaRAG, DRAGIN, and 4+ RAG methods on 5 datasets while cutting inference latency by **18%**.
- Automated **end-to-end** data processing pipelines to **standardize 6** major academic datasets for experimentation.

EEG2TEXT, Undergraduate Research Assistant

February 2025 – June 2025 | Blacksburg, VA

- Improved EEG-to-text **BLEU/ROUGE** scores **~5%** through multimodal pipeline optimization.
- Refactored and optimized **2000+** lines of code into modular components for improved readability and maintainability.
- Resolved **5+** data and pipeline issues by developing validation scripts and debugging multimodal EEG processing.

PROJECTS

OneRep – ML Exercise Recognition & Workout Trainer | *Swift, WatchKit, Core ML, PyTorch* | [GitHub](#) Sept. 2026

- Trained a PyTorch **temporal CNN** on **250K+** IMU samples, achieving **~91%** accuracy across **11** strength exercises.
- Improved classification F1 **~14%** using sliding windows, normalization, and time-series augmentation.
- Converted models to **Core ML** for **<40 ms** on-device inference while reducing model size by **~65%**.

RF Propagation & Ray Tracing Simulator | *Unreal Engine, NVIDIA Sionna, C++, TensorFlow*

June 2026

- Built a low-latency **pybind11** bridge embedding Python within **UE5 C++** for real-time **NVIDIA Sionna** ray tracing
- Offloaded ray-tracing workloads to **background threads** to prevent **UE5** game-thread blocking.
- Converted UE5 static-mesh buffers into **PLY** geometry for real-time Sionna scene ingestion.
- Achieved a stable **18-20** path solves per second by streaming multi-path solver polyline results to UE5 Blueprints.

BioSegment - Mitochondria and Cristae Segmentation Models | *PyTorch, ReactJS, Flask* | [GitHub](#)

Feb. 2026

- Built a **PyTorch** pipeline for **mitochondria/cristae segmentation** from EM images, cutting manual analysis by **95%**.
- Designed a human-in-the-loop labeling workflow, letting researchers validate and correct segmentation masks.
- Achieved **>0.8 IoU** using **Gaussian-blur** augmentation and **early stopping** to improve fidelity on noisy EM images.
- Deployed the **React/Flask** app to VT researchers, enabling **70%** more data labeling in the same timeframe.

Distributed Product Data Ingestion System | *AWS EC2, Flask, Selenium, Pandas, OpenAI API* | [GitHub](#)

Jan. 2026

- Deployed a data pipeline on **AWS EC2** to **automate** ingestion and **normalization** of e-commerce product data.
- Leveraged an **LLM** to synthesize unstructured HTML into schema-validated JSON metadata objects with Pydantic.
- Engineered fault-tolerant ingestion with retries, idempotency, **dead-letter recovery**, and **Prometheus metrics**.
- Benchmarked the parser across **150,000** fixture events with 100% success at approximately **3,813** events/second.

SKILLS

Programming languages: Python • JavaScript • C • C++ • Java • SQL • HTML • CSS • R • MATLAB

AI/ML: PyTorch • TensorFlow • vLLM • LangChain • Hugging Face • Scikit-Learn • ChromaDB • Computer Vision

Cloud & Tools: AWS • Docker • Kubernetes • React • Node.js • Flask • ElasticSearch • Linux • Git • CI/CD • Agile